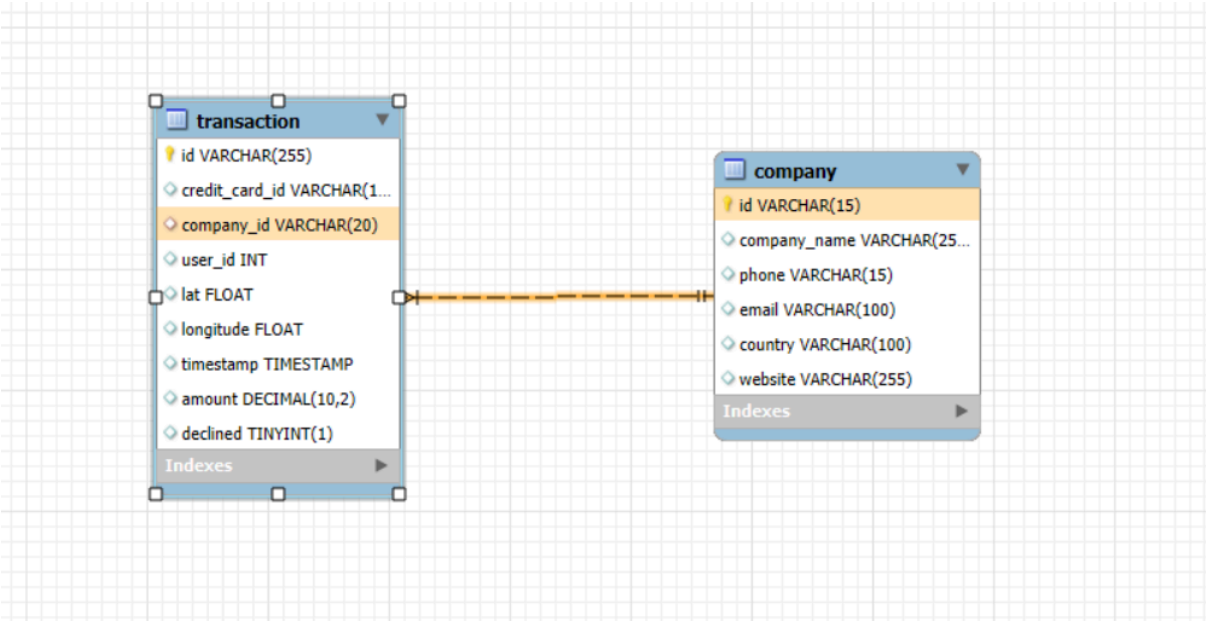
SPRINT 2
Ejercicio 1

La base de datos está formada por dos tablas, la tabla transaction que tiene la información de las transacciones y ventas con variables principales como: id (clave primaria), company_id (clave foránea), monto y fecha y la tabla company que tiene información sobre las empresas que realizan esas transacciones, tiene como variables principales id (llave primaria), company_name, country.

Se puede observar la relación en el diagrama, siendo en este caso 1 a muchos (1:N). Donde una empresa puede tener muchas transacciones pero una transacción pertenece a una única empresa .



Ejercicio 2 :
NIVELL 1
Utilitzant JOIN realitzaràs les següents consultes:

#Llistat dels països que estan generant vendes.

```
/*NIVELL 1
Exercici 2
Utilitzant JOIN realitzaràs les següents consultes:
*/

#Llistat dels països que estan generant vendes.
SELECT distinct country
FROM company c
JOIN transaction t
ON c.id=t.company_id
WHERE declined !=1;


#Des de quants països es generen les vendes.
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country
Germany
Australia
United States
New Zealand
Norway
United Kingdom

Result 57 | Read On

Output

Action Output

#	Time	Action	Message	Duration / Fetch
59	12:34:44	SELECT c.id,company_name, COUNT(*) as Transactions, CASE WHEN COUNT(*)> 400 THEN "Superior a 4...	100 row(s) returned	0.125 sec / 0.000 sec
60	12:34:50	SELECT c.id,company_name, COUNT(*) as Transactions, CASE WHEN COUNT(*)> 400 THEN "Superior a 4...	100 row(s) returned	0.110 sec / 0.000 sec
61	12:48:51	SELECT distinct country FROM company c JOIN transaction t ON c.id=t.company_id WHERE declined !=1	15 row(s) returned	0.016 sec / 0.000 sec

Des de quants països es generen les vendes.

```
#Des de quants països es generen les vendes.
SELECT count(distinct country) as Cantidad_Paises
FROM company c
JOIN transaction t
ON c.id=t.company_id
WHERE declined !=1;


#Identifica la companyia amb la mitjana més gran de vendes.
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Cantidad_Paises
15

Result 58 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
60	12:34:50	SELECT c.id,company_name, COUNT(*) as Transactions, CASE WHEN COUNT(*)> 400 THEN "Superior a 4...	100 row(s) returned	0.110 sec / 0.000 sec
61	12:48:51	SELECT distinct country FROM company c JOIN transaction t ON c.id=t.company_id WHERE declined !=1	15 row(s) returned	0.016 sec / 0.000 sec
62	12:49:45	SELECT count(distinct country) as Cantidad_Paises FROM company c JOIN transaction t ON c.id=t.company...	1 row(s) returned	0.219 sec / 0.000 sec

Identifica la companyia amb la mitjana més gran de vendes.

```
#Identifica la companyia amb la mitjana més gran de vendes.
SELECT c.company_name, round(AVG(amount),2) as Media_Ventas
FROM company c
JOIN transaction t
ON c.id=t.company_id
WHERE declined !=1
GROUP BY c.company_name
ORDER BY Media_Ventas DESC
limit 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	Media_Ventas
Ac Fermentum Incorporated	284.91

Result 59

Output

Action Output

#	Time	Action	Message	Duration / Fetch
61	12:48:51	SELECT distinct country FROM company c JOIN transaction t ON c.id=t.company_id WHERE declined !=1	15 row(s) returned	0.016 sec / 0.000 sec
62	12:49:45	SELECT count(distinct country) as Cantidad_Paises FROM company c JOIN transaction t ON c.id=t.company...	1 row(s) returned	0.219 sec / 0.000 sec
63	12:50:15	SELECT c.company_name, round(AVG(amount),2) as Media_Ventas FROM company c JOIN transaction t O...	1 row(s) returned	0.265 sec / 0.000 sec

Mostra totes les transaccions realitzades per empreses d'Alemanya.

```
/*Exercici 3
Utilitzant només subconsultes (sense utilitzar JOIN):
*/

#Mostra totes les transaccions realitzades per empreses d'Alemanya.
SELECT *
FROM transaction
where company_id in (
        SELECT id
        FROM company
        where country="Germany"
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
00138D3B-206D-4C03-94B7-63A2676EB9B4	CcS-4899	b-2222	318	41.3781	12.447	2020-03-25 10:43:43	426.36	0
0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9	CcS-5070	b-2222	489	41.3814	2.18176	2020-12-17 18:15:37	316.90	0
00201A11-2E62-44C4-941D-198FC8DB77F0	CcU-3512	b-2222	193	55.5704	-3.65129	2021-01-22 23:44:27	453.04	0
00235618-0A5C-4D49-9DCB-B3A9405D8923	CcS-8137	b-2222	3556	59.8421	18.729	2020-09-09 15:43:19	263.14	0
005A5A7B-1F1A-4B6C-9B15-1625A78C9C38	CcS-8998	b-2222	4417	41.1591	-8.63905	2024-05-15 09:10:11	442.01	0
00687139-48B2-4FFA-8F73-B20376F04AB4	CcS-4870	b-2222	289	51.1966	10.4669	2019-03-09 19:37:49	524.84	0

transaction 60

Output

Action Output

#	Time	Action	Message	Duration / Fetch
62	12:49:45	SELECT count(distinct country) as Cantidad_Paises FROM company c JOIN transaction t ON c.id=t.company...	1 row(s) returned	0.219 sec / 0.000 sec
63	12:50:15	SELECT c.company_name, round(AVG(amount),2) as Media_Ventas FROM company c JOIN transaction t O...	1 row(s) returned	0.265 sec / 0.000 sec
64	12:50:54	SELECT * FROM transaction where company_id in (SELECT id FROM company where country="Germany")	13291 row(s) returned	0.000 sec / 0.047 sec

Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

```
#Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.
SELECT *
FROM company c
WHERE EXISTS (
    SELECT 1
    FROM transaction t
    WHERE c.id=t.company_id
    AND amount>(
        SELECT AVG(amount) as promedio_monto
        FROM transaction
) AND declined !=1
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	company_name	phone	email	country	website
b-2222	Ac Fermentum Incorporated	06 85 56 52 33	donec.porttitor.tellus@yahoo.net	Germany	https://instagram.com/site
b-2226	Magna A Neque Industries	04 14 44 64 62	risus.donec.nibh@icloud.org	Australia	https://whatsapp.com/group/9
b-2230	Fusce Corp.	08 14 97 58 85	risus@protonmail.edu	United States	https://pinterest.com/sub/cars
b-2234	Convallis In Incorporated	06 66 57 29 50	mauris.ut@aol.couk	Germany	https://cnn.com/user/110
b-2238	Ante Iaculis Nec Foundation	08 23 04 99 53	sed.dictum.proin@outlook.ca	New Zealand	https://netflix.com/settings
b-2242	Donec Ltd	01 25 51 37 37	at.iaculis@hotmail.couk	Norway	https://nytimes.com/user/110

company 61

Output

Action Output

#	Time	Action	Message	Duration / Fetch
63	12:50:15	SELECT c.company_name, round(AVG(amount),2) as Media_Ventas FROM company c JOIN transaction t O...	1 row(s) returned	0.265 sec / 0.000 sec
64	12:50:54	SELECT * FROM transaction where company_id in (SELECT id FROM company where country="Germany")	13291 row(s) returned	0.000 sec / 0.047 sec
65	12:51:36	SELECT * FROM company c WHERE EXISTS (SELECT 1 FROM transaction t WHERE c.id=t.company_id ...	100 row(s) returned	0.079 sec / 0.000 sec

Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

No hay empresas sin transacciones

```
#Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.
#No hay empresas sin transacciones
SELECT *
FROM company c
where id not in(
    SELECT company_id
    FROM transaction
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	company_name	phone	email	country	website
NULL	NULL	NULL	NULL	NULL	NULL

company 62

Output

Action Output

#	Time	Action	Message	Duration / Fetch
64	12:50:54	SELECT * FROM transaction where company_id in (SELECT id FROM company where country="Germany")	13291 row(s) returned	0.000 sec / 0.047 sec
65	12:51:36	SELECT * FROM company c WHERE EXISTS (SELECT 1 FROM transaction t WHERE c.id=t.company_id ...	100 row(s) returned	0.079 sec / 0.000 sec
66	12:52:17	SELECT * FROM company c where id not in(SELECT company_id FROM transaction)	0 row(s) returned	0.031 sec / 0.000 sec

Nivell 2
Exercici 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes.
Mostra la data(fecha) de cada transacció juntament amb el total de les vendes.

```
/*Nivell 2
Exercici 1
Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes.
Mostra la data(fecha) de cada transacció juntament amb el total de les vendes.*/

SELECT DATE(timestamp) as Fecha, SUM(amount) as Total_ventas
FROM transaction
WHERE declined !=1
group by DATE(timestamp)
order by sum(amount) desc
limit 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Fecha	Total_ventas
2022-12-13	14337.44
2019-11-18	13591.32
2023-02-20	13332.59
2017-12-20	13318.43
2019-03-18	12680.95

Result 63

Output

Action Output

#	Time	Action	Message	Duration / Fetch
65	12:51:36	SELECT * FROM company c WHERE EXISTS (SELECT 1 FROM transaction t WHERE c.id=t.company_id ...	100 row(s) returned	0.079 sec / 0.000 sec
66	12:52:17	SELECT * FROM company c where id not in(SELECT company_id FROM transaction)	0 row(s) returned	0.031 sec / 0.000 sec
67	12:53:00	SELECT DATE(timestamp) as Fecha, SUM(amount) as Total_ventas FROM transaction WHERE declined !=1...	5 row(s) returned	0.078 sec / 0.000 sec

Exercici 2
Quina és la mitjana de vendes per país? Presenta els resultats ordenats de major a menor mitjà.

```
/*
Exercici 2
Quina és la mitjana de vendes per país? Presenta els resultats ordenats de major a menor mitjà.*/

SELECT country, ROUND(AVG(amount),2) as Media_ventas
FROM company c
JOIN transaction t
ON c.id=t.company_id
WHERE declined !=1
GROUP BY country
ORDER BY AVG(amount) DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	Media_ventas
Australia	265.54
United States	264.42
Belgium	260.97
Germany	260.83
Ireland	260.39
Spain	260.28

Result 64

Output

Action Output

#	Time	Action	Message	Duration / Fetch
66	12:52:17	SELECT * FROM company c where id not in(SELECT company_id FROM transaction)	0 row(s) returned	0.031 sec / 0.000 sec
67	12:53:00	SELECT DATE(timestamp) as Fecha, SUM(amount) as Total_ventas FROM transaction WHERE declined !=1...	5 row(s) returned	0.078 sec / 0.000 sec
68	12:53:35	SELECT country, ROUND(AVG(amount),2) as Media_ventas FROM company c JOIN transaction t ON c.id=t.c...	15 row(s) returned	0.266 sec / 0.000 sec

Exercici 3

En la teva empresa, es planteja un nou projecte per a llançar algunes campanyes publicitàries per a fer competència a la companyia "Non Institute".

Per a això, et demanen la llista de totes les transaccions realitzades per empreses que estan situades en el mateix país que aquesta companyia.

Mostra el llistat aplicant JOIN i subconsultes.

.

```
Exercici 3
En la teva empresa, es planteja un nou projecte per a llançar algunes campanyes publicitàries per a fer competència a la companyia "Non Institute".
 Per a això, et demanen la llista de totes les transaccions realitzades per empreses que estan situades en el mateix país que aquesta companyia.*/
#Mostra el llistat aplicant JOIN i subconsultes.
SELECT *
FROM company c
JOIN transaction t
ON c.id=t.company_id
WHERE country = (
    SELECT country
    FROM company
    where company_name="Non Institute"
) AND company_name!="Non institute"
;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

id	company_name	phone	email	country	website	id	credit_card_id	company_id	user_id	lat	longitude	timestamp
b-2246	Sed Nunc Ltd	02 62 64 73 48	nibh@yahoo.org	United Kingdom	https://cnn.com/one	008629B4-C9A9-406C-A3D2-71FDA47BC546	CcS-7063	b-2246	2482	45.7666	4.83048	2015-07-30 12:12:42
b-2246	Sed Nunc Ltd	02 62 64 73 48	nibh@yahoo.org	United Kingdom	https://cnn.com/one	00B72BA4-54A3-4B8E-B13F-2D57535AA17A	CcS-8475	b-2246	3894	55.6212	-3.7546	2017-10-26 22:08:26
b-2246	Sed Nunc Ltd	02 62 64 73 48	nibh@yahoo.org	United Kingdom	https://cnn.com/one	01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C	CcS-8700	b-2246	4119	55.856	-3.15783	2018-01-27 13:44:36
b-2246	Sed Nunc Ltd	02 62 64 73 48	nibh@yahoo.org	United Kingdom	https://cnn.com/one	023FFCE8-E618-4938-BF56-C8DF80540ADD	CcS-7816	b-2246	3235	46.3568	1.82755	2016-12-19 11:53:45
b-2246	Sed Nunc Ltd	02 62 64 73 48	nibh@yahoo.org	United Kingdom	https://cnn.com/one	02683BFB-FF91-4564-957B-D6F1662AB7C5	CcS-9471	b-2246	4890	42.1332	12.396	2017-01-10 21:09:29

Result 65 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
67	12:53:00	SELECT DATE(timestamp) as Fecha, SUM(amount) as Total_ventas FROM transaction WHERE declined !=1...	5 row(s) returned	0.078 sec / 0.000 sec
68	12:53:35	SELECT country, ROUND(AVG(amount),2) as Media_ventas FROM company c JOIN transaction t ON c.id=t.c...	15 row(s) returned	0.266 sec / 0.000 sec
69	12:54:22	SELECT * FROM company c JOIN transaction t ON c.id=t.company_id WHERE country = (SELECT country F...	12233 row(s) returned	0.000 sec / 0.047 sec

Mostra el llistat aplicant solament subconsultes

```
#Mostra el llistat aplicant solament subconsultes
SELECT *
FROM transaction
WHERE company_id in (
    SELECT id
    FROM company
    WHERE country in (
        SELECT country
        FROM company
        WHERE company_name="Non Institute"
) AND company_name!="Non Institute"
);
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

id	credit_card_id	company_id	user_id	lat	longitude	timestamp	amount	declined
0045F54B-D6D1-4582-9C3C-F1DAA46440D5	CcS-7661	b-2574	3080	48.8648	2.34847	2019-02-07 03:27:04	269.00	0
008ED95E-91D4-478B-9552-B935BBA32282	CcS-5727	b-2574	1146	51.2171	10.0583	2021-12-14 08:17:51	253.39	0
009B007D-CC8C-4EA0-AB8B-9036FA5F0CBC	CcS-8099	b-2574	3518	54.9513	-3.26943	2022-05-13 14:49:26	172.78	0
00F6558F-9660-44B1-9AD8-42EB09438137	CcS-7135	b-2574	2554	41.4135	12.665	2024-05-22 23:51:45	53.01	0
01104345-CF83-4594-9525-8BC00D0EA7EE	CcS-9414	b-2574	4833	45.7547	4.84544	2019-09-19 21:52:32	30.14	0
01609352-3871-456A-9F3A-64DC9568B6B5	CcS-6343	b-2574	1762	51.5161	19.4639	2021-08-29 16:42:36	417.47	0

transaction 66 | Apply

Output

Action Output

#	Time	Action	Message	Duration / Fetch
68	12:53:35	SELECT country, ROUND(AVG(amount),2) as Media_ventas FROM company c JOIN transaction t ON c.id=t.c...	15 row(s) returned	0.266 sec / 0.000 sec
69	12:54:22	SELECT * FROM company c JOIN transaction t ON c.id=t.company_id WHERE country = (SELECT country F...	12233 row(s) returned	0.000 sec / 0.047 sec
70	12:55:01	SELECT * FROM transaction WHERE company_id in (SELECT id FROM company WHERE country in (SE...	12233 row(s) returned	0.016 sec / 0.093 sec

Nivell 3

Exercici 1

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i
en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

```
/*Nivell 3
Exercici 1
Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i
en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.*/

SELECT country, company_name, phone, DATE(t.timestamp) as Date ,amount
FROM company c
JOIN transaction t
ON c.id=t.company_id
where amount BETWEEN 350 AND 400
AND DATE(t.timestamp) in ("2015-04-29","2018-07-20","2024-03-13") #sino ponemos las fechas entre comillas serian operaciones matematicas
ORDER BY amount DESC;

/*Exercici 2
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country	company_name	phone	Date	amount
Germany	Aliquam PC	01 45 73 52 16	2024-03-13	399.84
United States	Auctor Mauris Vel LLP	08 09 28 74 14	2018-07-20	399.51
Italy	At Pede Corp.	06 14 48 33 15	2015-04-29	390.69
Germany	Aliquam PC	01 45 73 52 16	2024-03-13	388.29
United Kingdom	Orci Adipiscing Limited	03 18 00 77 81	2018-07-20	373.71
New Zealand	Fringilla LLC	08 29 15 93 57	2015-04-29	367.62

Result 67

Output

Action Output

#	Time	Action	Message	Duration / Fetch
69	12:54:22	SELECT * FROM company c JOIN transaction t ON c.id=t.company_id WHERE country = (SELECT country F...	12233 row(s) returned	0.000 sec / 0.047 sec
70	12:55:01	SELECT * FROM transaction WHERE company_id in (SELECT id FROM company WHERE country in (SE...	12233 row(s) returned	0.016 sec / 0.093 sec
71	12:55:33	SELECT country, company_name, phone, DATE(t.timestamp) as Date ,amount FROM company c JOIN trans...	8 row(s) returned	0.031 sec / 0.000 sec

Exercici 2

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre
la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis
si tenen més de 400 transaccions o menys.

```
/*Exercici 2
Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre
la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis
 si tenen més de 400 transaccions o menys.*/
SELECT c.id,company_name, COUNT(*) as Transactions,
CASE
    WHEN COUNT(*)> 400 THEN "Superior a 400"
    ELSE "Menor a 400"
END as Classification
FROM company c
JOIN transaction t
ON c.id=t.company_id
GROUP BY c.id,company_name
order by Transactions DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

id	company_name	Transactions	Classification
b-2222	Ac Fermentum Incorporated	2401	Superior a 400
b-2302	Nunc Interdum Incorporated	1599	Superior a 400
b-2330	Donec Fringilla PC	1593	Superior a 400
b-2366	Mauris Institute	1586	Superior a 400
b-2614	Rutrum Non Inc.	1585	Superior a 400
b-2350	Aliquet Vel Vulputate Incorporated	1583	Superior a 400

Result 68 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
70	12:55:01	SELECT * FROM transaction WHERE company_id in (SELECT id FROM company WHERE country in (SE...	12233 row(s) returned	0.016 sec / 0.093 sec
71	12:55:33	SELECT country, company_name, phone, DATE(t.timestamp) as Date ,amount FROM company c JOIN trans...	8 row(s) returned	0.031 sec / 0.000 sec
72	12:56:04	SELECT c.id,company_name, COUNT(*) as Transactions, CASE WHEN COUNT(*)> 400 THEN "Superior a 4...	100 row(s) returned	0.093 sec / 0.000 sec